Replaying the Keystroke: Session Reconstruction for Instructor Sense-Making in Large Programming Courses

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In Proceedings of the ACM Technical Symposium on Computer Science Education (SIGCSE TS '24), pp. 512-518.

Abstract

Instructors of large programming courses see students' final submissions but rarely the process that produced them. We built SessionReel, a tool that reconstructs a student's editing session from keystroke logs and plays it back at adjustable speed with run and error markers. In a within-subjects study, 9 instructors and 14 teaching assistants examined 40 sessions and were asked to locate the moment a student became stuck. With final code alone, participants located the stuck point within 60 seconds in 41% of sessions; with replay, 77%. Replay cost a median 3.8 minutes per session. We report which visual markers helped and argue that replay is most useful as a targeted tool for sessions already flagged by simpler signals.

1 Introduction

A CS1 course with 600 students produces tens of thousands of submissions per term. Autograders tell instructors whether code works; they do not tell them how a student got there. Process data such as keystrokes, runs and error messages has been collected for over a decade, but it is usually aggregated into dashboards of counts. Counts flatten the moments that matter to a teacher: the first failed run, the long pause, the deletion of a working function.

We ask whether replaying a session, much like scrubbing through a screen recording, helps instructors make sense of struggle, and at what cost in time.

2 SessionReel

SessionReel reconstructs file state at every keystroke from an append-only event log (insert, delete, paste, run, test result). The player shows the editor on the left and a timeline on the right. The timeline carries three marker types: red ticks for runs that raised an exception, green ticks for passing test runs, and grey bands for idle gaps longer than 45 seconds. Playback runs at 1x to 32x; idle gaps can be skipped automatically.

Reconstruction is exact for typed edits. Pastes are reconstructed as single events, which preserves their size but not their origin.

3 Study

Participants (9 instructors, 14 TAs) each reviewed 20 sessions in two blocks: final code plus error log (baseline), and SessionReel (replay). Sessions came from a single assignment on list processing and were sampled to include 30 with an expert-annotated "stuck point" and 10 without one. Block order was counterbalanced.

Condition	Stuck point found (<= 60 s)	False alarms	Median time / session
Final code + error log	41%	0.22	1.1 min

Replay, no markers	63%	0.31	4.6 min
Replay with markers	77%	0.18	3.8 min

Markers mattered more than playback itself. Without markers, participants scrubbed back and forth and reported "watching typing" as tedious. Idle-gap bands were cited most often as useful (17 of 23 participants), followed by error ticks (14 of 23).

4 Findings

Instructors used replay to answer three questions: where did the student stop making progress, what did they try next, and did they seem to understand the error message. The third was the hardest; participants disagreed on it in 38% of sessions. TAs, who had held office hours that term, were faster than instructors (median 3.1 vs 4.4 minutes) but no more accurate.

Several participants said replay felt intrusive. One instructor noted that she would not want students to feel watched while they were "allowed to be confused". We return to this in the discussion.

5 Discussion

Replay is expensive. At 3.8 minutes per session, reviewing every session in a 600-student course is out of reach. We therefore recommend replay as a second stage: a cheap signal (repeated identical errors, idle gaps after a failed run) selects candidate sessions, and an instructor or TA uses replay to decide whether to intervene.

Students should know that sessions can be replayed and by whom. In our deployment, consent forms described replay explicitly, and 4% of students opted out.

6 Limitations

One assignment, one institution, expert-annotated stuck points that are themselves a judgment. We did not measure whether replay changed what instructors did next.

References

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